THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **A Tale of Three Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 80°	Share 180° in the ratio.
2	**(c)** 70°	Exterior angle equals the sum of the two base angles.
3	**(c)** 75°	Exterior angle $= \angle A + \angle C$.
4	**(c)** 7	Count whole numbers strictly between the difference and the sum.
5	**(d)** 70°	Substitute d.
6	**(d)** SAS	Check which parts are given and where they sit.
7	**(b)** 24	Add the three sides.
8	**(a)** 110°	The other acute angle is $90^\circ - a$; the exterior angle is supplementary to it.
9	**(d)** 90°	Share 180° in the ratio.
10	**(b)** 100°	Exterior angle equals the sum of the two base angles.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Case 1: 6 is an equal side, base $= 26 - 2 \times 6 = 14$; not valid, since $6 + 6 = 12$ is not greater than 14; Case 2: 6 is the base, equal sides $= (26 - 6) \div 2 = 10$; valid (1); Sides (cm): $10, 10, 6$ (1).	1+1
12	Exterior = sum of the interior opposite angles; $4x - 10 = 3x + 30$ so $x = 40$ (1); Exterior angle $= 150^\circ$ (1).	1+1
13	$\angle B = 180^\circ - 110^\circ = 70^\circ$; $\angle C = \angle B = 70^\circ$ (equal sides) (1); $\angle A = 180^\circ - 2(70^\circ) = 40^\circ$ (1).	1+1
14	Case 1: 10 is an equal side, base $= 22 - 2 \times 10 = 2$; valid; Case 2: 10 is the base, equal sides $= (22 - 10) \div 2 = 6$; valid (1); Sides (cm): $10, 10, 2$ or $6, 6, 10$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	One part $= 180^\circ \div 9 = 20^\circ$; Angles 40°, 60°, 80° (1); Exterior angle $= 180^\circ - 80^\circ$ (1); $= 100^\circ$ (1).	1+1+1
16	Equal sides, so $\angle Q = \angle R$; $3x - 10 = x + 6$ so $x = 8$ (1); $\angle Q = \angle R = 14^\circ$; $\angle P = 180^\circ - 28^\circ = 152^\circ$ (1); Exterior angle at P $= 28^\circ$ (1).	1+1+1
17	Corresponding sides of congruent triangles are equal, so $AB = DE$; $5x - 2x = 18 - 3$, so $3x = 15$ and $x = 5$ (1); $DE = AB = 28$ cm; Perimeter $= AB + BC + AC = 28 + 7 + 9$ (1); $= 44$ cm (1).	1+1+1
	OR: One part $= 180^\circ \div 6 = 30^\circ$; Angles 30°, 60°, 90° (1); Exterior angle $= 180^\circ - 30^\circ$ (1); $= 150^\circ$ (1).	1+1+1
18	Equal sides, so $\angle Q = \angle R$; $3x - 20 = x + 10$ so $x = 15$ (1); $\angle Q = \angle R = 25^\circ$; $\angle P = 180^\circ - 50^\circ = 130^\circ$ (1); Exterior angle at P $= 50^\circ$ (1).	1+1+1
	OR: Corresponding sides of congruent triangles are equal, so $AB = DE$; $4x - 2x = 23 - 1$, so $2x = 22$ and $x = 11$ (1); $DE = AB = 45$ cm; Perimeter $= AB + BC + AC = 45 + 6 + 8$ (1); $= 59$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Equal sides: $2x + 1 = 3x - 4$, so $x = 5$ (1); Each equal side $= 11$ cm (1); Base $= 5 + 3 = 8$ cm (1); Perimeter $= 11 + 11 + 8 = 30$ cm (1); Existence: $11 + 8 = 19 > 11$ and $11 + 11 = 22 > 8$ (1).	1+1+1+1+1
	OR: $2, 4, 6$: 2+4=6 not > 6, no triangle; $2, 4, 7$: 2+4=6 not > 7, no triangle; $2, 6, 7$: 2+6=8 > 7, triangle; $4, 6, 7$: 4+6=10 > 7, triangle (2); Number of triangles $= 2$ (3).	2+3
20	Equal sides: $2x + 2 = 3x - 3$, so $x = 5$ (1); Each equal side $= 12$ cm (1); Base $= 5 + 6 = 11$ cm (1); Perimeter $= 12 + 12 + 11 = 35$ cm (1); Existence: $12 + 11 = 23 > 12$ and $12 + 12 = 24 > 11$ (1).	1+1+1+1+1
	OR: $3, 5, 8$: 3+5=8 not > 8, no triangle; $3, 5, 9$: 3+5=8 not > 9, no triangle; $3, 8, 9$: 3+8=11 > 9, triangle; $5, 8, 9$: 5+8=13 > 9, triangle (2); Number of triangles $= 2$ (3).	2+3

Mistakes to avoid:

- Checking only one pair of sides in the triangle inequality; all three conditions must hold (the smallest pair is the critical one).

- Saying a triangle is possible when two sides add up exactly to the third; the sum must be *greater*.
- Using 90° or 360° instead of 180° for the angle sum of a triangle.
- Equating the exterior angle to the adjacent interior angle instead of the sum of the two interior opposite angles.
- In an isosceles triangle, taking the angles opposite the unequal side as equal.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **A Tale of Three Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 65°	Exterior angle $= \angle A + \angle C$.
2	**(b)** 11	Count whole numbers strictly between the difference and the sum.
3	**(d)** 80°	Substitute d.
4	**(c)** ASA	Check which parts are given and where they sit.
5	**(a)** 34	Add the three sides.
6	**(c)** 125°	The other acute angle is $90^\circ - a$; the exterior angle is supplementary to it.
7	**(d)** 75°	Share 180° in the ratio.
8	**(c)** 124°	Exterior angle equals the sum of the two base angles.
9	**(d)** 75°	Exterior angle $= \angle A + \angle C$.
10	**(d)** 5	Count whole numbers strictly between the difference and the sum.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Exterior = sum of the interior opposite angles; $4x - 6 = 3x + 34$ so $x = 40$ (1); Exterior angle $= 154^\circ$ (1).	1+1
12	$\angle B = 180^\circ - 130^\circ = 50^\circ$; $\angle C = \angle B = 50^\circ$ (equal sides) (1); $\angle A = 180^\circ - 2(50^\circ) = 80^\circ$ (1).	1+1
13	Case 1: 8 is an equal side, base $= 36 - 2 \times 8 = 20$; not valid, since $8 + 8 = 16$ is not greater than 20; Case 2: 8 is the base, equal sides $= (36 - 8) \div 2 = 14$; valid (1); Sides (cm): $14, 14, 8$ (1).	1+1
14	Exterior = sum of the interior opposite angles; $4x - 8 = 3x + 50$ so $x = 58$ (1); Exterior angle $= 224^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	One part $= 180^\circ \div 12 = 15^\circ$; Angles 45°, 60°, 75° (1); Exterior angle $= 180^\circ - 75^\circ$ (1); $= 105^\circ$ (1).	1+1+1
16	Equal sides, so $\angle Q = \angle R$; $3x - 6 = x + 12$ so $x = 9$ (1); $\angle Q = \angle R = 21^\circ$; $\angle P = 180^\circ - 42^\circ = 138^\circ$ (1); Exterior angle at P $= 42^\circ$ (1).	1+1+1
17	Corresponding sides of congruent triangles are equal, so $AB = DE$; $7x - 3x = 30 - 2$, so $4x = 28$ and $x = 7$ (1); $DE = AB = 51$ cm; Perimeter $= AB + BC + AC = 51 + 5 + 11$ (1); $= 67$ cm (1).	1+1+1
	OR: One part $= 180^\circ \div 12 = 15^\circ$; Angles 15°, 60°, 105° (1); Exterior angle $= 180^\circ - 15^\circ$ (1); $= 165^\circ$ (1).	1+1+1
18	Equal sides, so $\angle Q = \angle R$; $3x - 14 = x + 6$ so $x = 10$ (1); $\angle Q = \angle R = 16^\circ$; $\angle P = 180^\circ - 32^\circ = 148^\circ$ (1); Exterior angle at P $= 32^\circ$ (1).	1+1+1
	OR: Corresponding sides of congruent triangles are equal, so $AB = DE$; $6x - 2x = 24 - 4$, so $4x = 20$ and $x = 5$ (1); $DE = AB = 34$ cm; Perimeter $= AB + BC + AC = 34 + 8 + 10$ (1); $= 52$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Equal sides: $2x + 3 = 3x - 5$, so $x = 8$ (1); Each equal side $= 19$ cm (1); Base $= 8 + 2 = 10$ cm (1); Perimeter $= 19 + 19 + 10 = 48$ cm (1); Existence: $19 + 10 = 29 > 19$ and $19 + 19 = 38 > 10$ (1).	1+1+1+1+1
	OR: $4, 5, 9$: 4+5=9 not > 9, no triangle; $4, 5, 10$: 4+5=9 not > 10, no triangle; $4, 9, 10$: 4+9=13 > 10, triangle; $5, 9, 10$: 5+9=14 > 10, triangle (2); Number of triangles $= 2$ (3).	2+3
20	Equal sides: $2x + 4 = 3x - 2$, so $x = 6$ (1); Each equal side $= 16$ cm (1); Base $= 6 + 7 = 13$ cm (1); Perimeter $= 16 + 16 + 13 = 45$ cm (1); Existence: $16 + 13 = 29 > 16$ and $16 + 16 = 32 > 13$ (1).	1+1+1+1+1
	OR: $2, 3, 7$: 2+3=5 not > 7, no triangle; $2, 3, 8$: 2+3=5 not > 8, no triangle; $2, 7, 8$: 2+7=9 > 8, triangle; $3, 7, 8$: 3+7=10 > 8, triangle (2); Number of triangles $= 2$ (3).	2+3

Mistakes to avoid:

- Checking only one pair of sides in the triangle inequality; all three conditions must hold (the smallest pair is the critical one).
- Saying a triangle is possible when two sides add up exactly to the third; the sum must be *greater*.
- Using 90° or 360° instead of 180° for the angle sum of a triangle.

- Equating the exterior angle to the adjacent interior angle instead of the sum of the two interior opposite angles.
- In an isosceles triangle, taking the angles opposite the unequal side as equal.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **A Tale of Three Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 85°	Substitute d.
2	**(d)** SSS	Check which parts are given and where they sit.
3	**(b)** 38	Add the three sides.
4	**(c)** 130°	The other acute angle is $90^\circ - a$; the exterior angle is supplementary to it.
5	**(a)** 110°	Share 180° in the ratio.
6	**(b)** 140°	Exterior angle equals the sum of the two base angles.
7	**(b)** 65°	Exterior angle $= \angle A + \angle C$.
8	**(a)** 9	Count whole numbers strictly between the difference and the sum.
9	**(a)** 75°	Substitute d.
10	**(c)** not necessarily congruent	Check which parts are given and where they sit.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\angle B = 180^\circ - 120^\circ = 60^\circ$; $\angle C = \angle B = 60^\circ$ (equal sides) (1); $\angle A = 180^\circ - 2(60^\circ) = 60^\circ$ (1).	1+1
12	Case 1: 4 is an equal side, base $= 18 - 2 \times 4 = 10$; not valid, since $4 + 4 = 8$ is not greater than 10; Case 2: 4 is the base, equal sides $= (18 - 4) \div 2 = 7$; valid (1); Sides (cm): $7, 7, 4$ (1).	1+1
13	Exterior = sum of the interior opposite angles; $4x - 4 = 3x + 34$ so $x = 38$ (1); Exterior angle $= 148^\circ$ (1).	1+1
14	$\angle B = 180^\circ - 140^\circ = 40^\circ$; $\angle C = \angle B = 40^\circ$ (equal sides) (1); $\angle A = 180^\circ - 2(40^\circ) = 100^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	One part $= 180^\circ \div 18 = 10^\circ$; Angles 20°, 50°, 110° (1); Exterior angle $= 180^\circ - 110^\circ$ (1); $= 70^\circ$ (1).	1+1+1
16	Equal sides, so $\angle Q = \angle R$; $3x - 12 = x + 8$ so $x = 10$ (1); $\angle Q = \angle R = 18^\circ$; $\angle P = 180^\circ - 36^\circ = 144^\circ$ (1); Exterior angle at P $= 36^\circ$ (1).	1+1+1
17	Corresponding sides of congruent triangles are equal, so $AB = DE$; $8x - 3x = 30 - 5$, so $5x = 25$ and $x = 5$ (1); $DE = AB = 45$ cm; Perimeter $= AB + BC + AC = 45 + 9 + 12$ (1); $= 66$ cm (1).	1+1+1
	OR: One part $= 180^\circ \div 4 = 45^\circ$; Angles 45°, 45°, 90° (1); Exterior angle $= 180^\circ - 45^\circ$ (1); $= 135^\circ$ (1).	1+1+1
18	Equal sides, so $\angle Q = \angle R$; $3x - 18 = x + 10$ so $x = 14$ (1); $\angle Q = \angle R = 24^\circ$; $\angle P = 180^\circ - 48^\circ = 132^\circ$ (1); Exterior angle at P $= 48^\circ$ (1).	1+1+1
	OR: Corresponding sides of congruent triangles are equal, so $AB = DE$; $9x - 4x = 33 - 3$, so $5x = 30$ and $x = 6$ (1); $DE = AB = 57$ cm; Perimeter $= AB + BC + AC = 57 + 10 + 7$ (1); $= 74$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Equal sides: $2x + 1 = 3x - 9$, so $x = 10$ (1); Each equal side $= 21$ cm (1); Base $= 10 + 5 = 15$ cm (1); Perimeter $= 21 + 21 + 15 = 57$ cm (1); Existence: $21 + 15 = 36 > 21$ and $21 + 21 = 42 > 15$ (1).	1+1+1+1+1
	OR: $5, 6, 11$: 5+6=11 not > 11, no triangle; $5, 6, 12$: 5+6=11 not > 12, no triangle; $5, 11, 12$: 5+11=16 > 12, triangle; $6, 11, 12$: 6+11=17 > 12, triangle (2); Number of triangles $= 2$ (3).	2+3
20	Equal sides: $2x + 5 = 3x - 3$, so $x = 8$ (1); Each equal side $= 21$ cm (1); Base $= 8 + 4 = 12$ cm (1); Perimeter $= 21 + 21 + 12 = 54$ cm (1); Existence: $21 + 12 = 33 > 21$ and $21 + 21 = 42 > 12$ (1).	1+1+1+1+1
	OR: $3, 4, 7$: 3+4=7 not > 7, no triangle; $3, 4, 12$: 3+4=7 not > 12, no triangle; $3, 7, 12$: 3+7=10 not > 12, no triangle; $4, 7, 12$: 4+7=11 not > 12, no triangle (2); Number of triangles $= 0$ (3).	2+3

Mistakes to avoid:

- Checking only one pair of sides in the triangle inequality; all three conditions must hold (the smallest pair is the critical one).
- Saying a triangle is possible when two sides add up exactly to the third; the sum must be *greater*.
- Using 90° or 360° instead of 180° for the angle sum of a triangle.

- Equating the exterior angle to the adjacent interior angle instead of the sum of the two interior opposite angles.
- In an isosceles triangle, taking the angles opposite the unequal side as equal.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND